

Muhammad Ekramah

Bioinformatics Graduate | Genomics and Computational Biology

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Portfolio: itsekramah.github.io

Academic Profile

Bioinformatics graduate with academic projects in genomic prediction, RNA-seq analysis and biological software, and internship experience in antimicrobial-resistance surveillance. Interested in quantitative genomics, systems biology and reproducible analysis of high-dimensional biological data. Seeking deeper statistical and computational training through Padua's MSc in Quantitative and Computational Biosciences.

Education

BS in Bioinformatics

Sep 2022 – Sep 2026

National Centre for Bioinformatics, Quaid-i-Azam University, Islamabad, Pakistan

Relevant coursework: Probability and Biostatistics; Bioinformatics Computing; Genomics; Functional Genomics; Systems Biology; Artificial Intelligence; Data Structures and Algorithms; Modelling and Simulation; Database Management Systems.

Selected Academic Projects

Genomic prediction of antimicrobial resistance

Repository

Bioinformatics Computing II | Oct–Dec 2025; revised analysis, 2026

- Analysed binary genomic unitigs from *Neisseria gonorrhoeae*; audited 3,786 metadata records and matched phenotype-specific samples for azithromycin and ciprofloxacin prediction.
- The revised notebook compares class-weighted logistic regression and random forest with a prevalence baseline using five-fold cross-validation grouped by identical unitig profiles; variance filtering and chi-square feature selection occur within training folds.
- Evaluated class imbalance, out-of-fold predictions, profile-group bootstrap intervals and descriptive geographic subgroup performance. Excluded cefixime from model comparison because only five isolates were resistant.
- Documented the limits of already GWAS-filtered inputs: internal performance may be optimistic, and predictive unitigs are not validated resistance mechanisms. The repository includes executed notebooks, input data, versioned tables and figures, and a reproducibility workflow.

RNA-seq: Notch2 conditional knockout in hippocampal stem cells

Repository

Genomics project | Feb–Jul 2025; count-level reanalysis, 2026

- Reanalysed public GSE116773 gene counts, aggregating 24 technical files into six biological RNA preparations: three wild-type and three Notch2 conditional-knockout samples.
- Used PyDESeq2 negative-binomial modelling, Wald tests and Benjamini–Hochberg correction; the saved analysis reports 121 genes at $FDR < 0.05$ among 21,531 retained genes (77 higher and 44 lower in knockout).
- Examined blind variance-stabilised PCA, sample distances and gene-level counts. Interpreted differences as associations, with small sample size and an unusual WT3 sample limiting inference.
- Preserved the earlier Bash/R workflow (FastQC, MultiQC, Trimmomatic, HISAT2, featureCounts and DESeq2) separately; its missing original count matrix prevents reproduction of the historical results.

Selected Academic Projects (continued)

PathoScope AI: viral genomics and transcriptomics prototype

Repository

Functional Genomics team project / Mar–Aug 2026

- Contributed to a team prototype with a FastAPI backend and Next.js interface. The project contribution notes identify my focus as workflow orchestration, FASTQ processing integration, PubMed retrieval and AI-assisted reporting.
- The implementation includes sequential workflow steps, job-status tracking, NCBI E-Utilities literature retrieval and caching, and wrappers for FastQC, fastp and SPAdes.
- The team code also includes six-frame ORF scanning and an exploratory expression module using log-CPM, Welch's tests and BH correction. External-tool fallbacks can generate simulated annotations or copy a reference sequence; these outputs are not evidence of successful alignment or assembly.

Experience

Research Intern

Jul–Aug 2025

National Institutes of Health (NIH), Islamabad, Pakistan

- Used R and the AMR package to analyse antimicrobial-resistance surveillance data, including wastewater data, and summarise susceptibility patterns.
- Gained exposure to PCR-based diagnostics, Oxford Nanopore MinION sequencing and bulk RNA-seq workflows, connecting data generation and quality control with downstream analysis.

Python Developer Intern

Jul–Sep 2024

Digital Empowerment Network | Remote

- Worked on Python backend services and REST APIs using FastAPI and PostgreSQL.
- Automated data cleaning and processing with Pandas and NumPy, and developed visualisation utilities for GitHub Classroom analytics in a Git-based team workflow.

Technical Skills

Programming	Python (Pandas, NumPy, scikit-learn), R, Bash, SQL
Quantitative analysis	Classification; grouped cross-validation; feature selection; bootstrap uncertainty; PCA; negative-binomial modelling; hypothesis testing; FDR correction
Genomics	Genomic unitig data; bulk RNA-seq; PyDESeq2; DESeq2; sequence QC, alignment and quantification workflows
Software and workflow	Git/GitHub; executable notebooks; dependency pinning; automated checks; FastAPI; PostgreSQL
Data resources	NCBI GEO/SRA and PubMed; Ensembl; familiarity with UniProt and Pfam through team-project integration

Research Interests

Statistical analysis of genomic and transcriptomic data; antimicrobial-resistance prediction and pathogen surveillance; systems biology; interpretable, reproducible computational research.